Project Brief

Release 1.0 – 26 November 2025

Overview

Student groups (“Companies”) will deliver a mission plan, systems, and interfaces required to locate a wilderness casualty and deliver a first aid kit in appropriate proximity. A demonstration and validation flight will be conducted in the presence of Flight Lab members and external partners (“the Client”).

The mission will be demonstrated at the UoB aerial robotics test facility (“Fenswood Wilderness”, Figure 1) and comprises challenges based on current Search and Rescue (SAR) research activities.

This document should be considered ‘live’, i.e. amendments may be issued in response to discussions between Client and Companies. The most up-to-date version will be published via Teams[1].

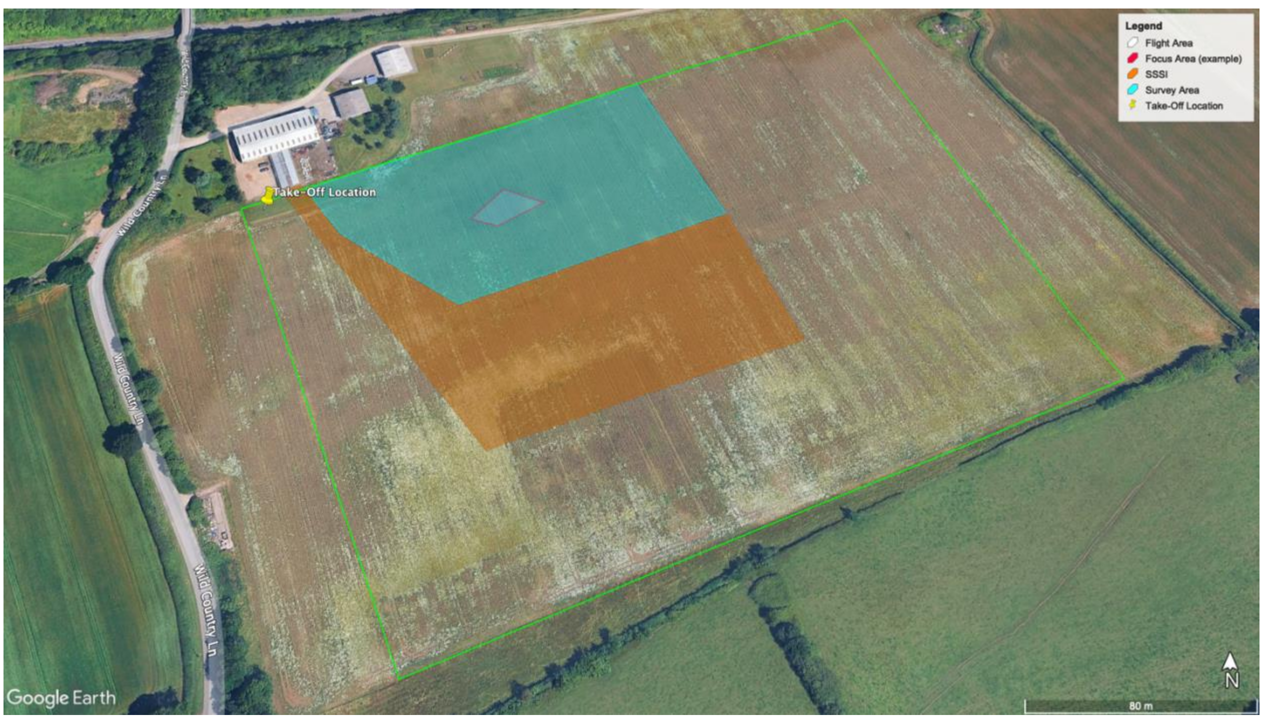

Figure 1: Fenswood Wilderness (.kml file available)

[1] For reassurance, no ‘surprise amendments’ are planned for this project. Amendments will be made in good faith in response to Company or other situational requirements.

Scenario

A hiker traversing the Fenswood Wilderness has been reported as lost. A description has been given of their height, clothing, and equipment, and a broad search region has been defined.

Mid-search the hiker activates a Personal Locator Beacon (PLB), giving a narrower region upon which to focus the search.

The search region contains a Site of Specific Scientific Interest (SSSI) where rare birds nest. This area is fenced off and inaccessible to hikers. The SSSI should not be overflown.

Requirements

R	Requirement	Note
R01	The drone shall only fly within the Flight Area.	1
R02	The drone shall not fly over the SSSI.	1
R03	The drone shall take off from a region within 5m of the defined Take-Off Location (TOL).	1
R04	The drone shall not exceed an altitude of 50m above TOL ground level.	1
R05	The drone shall undertake a search of the Search Area and identify potential items of lost person clothing and equipment ("items of interest").	1
R06	Upon PLB activation, which is expected to occur 5-15 minutes into the search, teams will receive coordinates of a Focus Area, defined by n lat/long coordinates, where $3 \leq n \leq 10$), upon which the drone should then focus its search.	
R07	The drone shall land within 10 metres but not within 5 metres of the casualty, deploy the provided first aid kit, then return to TOL.	
R07	The level of autonomy and interface format and functionality shall be determined and justified by the Company and approved by the Client.	
R08	All interfaces (RC, Ground Station, and any others) shall include 'Return to Home' (RTH) and 'Motor Cutoff' commands and other appropriate failsafes.	
R09	Latitude/longitude and images of detected items of interest and of the casualty shall be reported. Verification report appendix including relevant imagery, data, uncertainties, etc., with details agreed between Company and Client.	
R10	The operation must have a designated Company Pilot, who will be responsible for compliant operation of the activity. The Company Pilot shall be supervised by a Flight Lab Safety Pilot, who shall have ultimate responsibility and absolute authority over all operations. Additional responsibilities may be allocated as required at Company and Lab discretion.	
R11	All software and flight logs shall be delivered via a public GitHub repository linked from the Company Report, licensed under the MIT licence.	

1. Coordinates are available in AENGM0074.kml. Areas will not be visibly marked in any field trials, but teams should report any excursions/incursions verbally during trials and in post-flight reports, and flight logs may be examined to verify compliance. Use of automatic geofences is required.

Equipment

The Research Drone Platform will be provided preassembled. Companies may augment or modify elements with prior discussion with and approval of the Flight Lab team. The Platform shall be returned in its original state post-demonstration.

Item	**Description**	**Quantity**	**Specifications**
Hexsoon EDU-450	Drone platform	1	https://ardupilot.org/copter/docs/reference-frames-hexsoon-edu450.html
Cube Orange	Flight controller	1	https://docs.cubepilot.org/user-guides/autopilot/the-cube-module-overview
Here 3+	GPS receiver	1	https://docs.cubepilot.org/user-guides/here-3/here-3-manual
FrSky TW-Mini	RC receiver	1	https://www.frsky-rc.com/product/tw-mini/
Raspberry Pi 5 8GB	Computer	1	https://www.raspberrypi.com/products/raspberry-pi-5/
Raspberry Pi camera (global shutter)	Camera	1	https://www.raspberrypi.com/products/raspberry-pi-global-shutter-camera/
Pi camera lens (6mm wide angle)	Lens	1	https://thepihut.com/products/raspberry-pi-high-quality-camera-lens
Pi camera cable	Cable	1	
SanDisk Extreme PRO 128GB	Micro SD storage	2	https://shop.sandisk.com/en-gb/products/memory-cards/microsd-cards/sandisk-extreme-pro-uhs-i-microsd
FrSky Twin X14	RC transmitter	1	https://www.frsky-rc.com/product/twin-x14/
Tarot Payload Release Mechanism (double throw)	Actuator	1	http://tarotrc.com/Product/Detail.aspx?Lang=en&Id=fc95cb53-74ae-4673-8fcc-541ac2d91631
Plus relevant cables, mounts, housings, etc. as required.			

Imagery

Sample imagery will be provided, and Companies will have the opportunity to obtain their own datasets during trial flights.

Rescue dummy – simulated SAR casualty (will be attired differently)

Project Deliverables

D	Deliverable	Description	Due	%
D1	Platform validation flight	Familiarisation flights, manual and waypoint.	Wednesday wk14 In-person	Formative
D2	Preliminary Design Review presentation	Outline proposal including autonomy, timeline, technologies.	Wednesday wk15 In-person	Formative
D3	Trial flights	Integrated system demonstration, trial data capture	Wednesday wk19/20/21 In-person	Formative
D4	Demonstration flights	Full system demonstration, final data capture	Wednesday wk22 In-person	Formative
D5	Final Design Review presentation	Requirements verification, initial evaluation	Wednesday wk22 In-person	Formative
D6	Group Report	Process, outcomes, lessons learned	Thursday wk23 Group submission (peer weighted)	50%
D7	Individual Reflective Report	Summary of contribution and personal development	Thursday wk23 Individual submission	50%

- Week numbers as given at https://www.bristol.ac.uk/university/dates/calendar/.
- Dates subject to weather-dependent change.
- Blackboard (Bb) submission guidelines at https://www.bristol.ac.uk/digital-education/students/.
- Late penalties as detailed in §17.4 (modular programmes) at https://www.bristol.ac.uk/academic-quality/assessment/regulations-and-code-of-practice-for-taught-programmes/penalties/.
- Submissions will be checked for academic integrity and are subject to University regulations, information at https://www.bristol.ac.uk/digital-education/assessment-online/academic-integrity/.
- AI usage is permitted per guidelines at https://www.bristol.ac.uk/bilt/sharing-practice/guides/guidance-on-ai/using-ai-in-assessment/, with restrictions on:
    - Software development - Category 3: Selective
    - Reporting and presentation - Category 2: Minimal

D2 Preliminary Design Review (PDR) presentation (formative)

- In-person presentation with accompanying slides
- 15 min presentation + 10 min questions
- Each company member should present at least one slide each

Audience:

- MSc peers
- Flight Lab academics and postgrads

Demonstrate that:

- Your planned work should meet project requirements
- You know what you need to do and when
- You are working together effectively to meet your goals

You could include:

- An overview of your planned system
  - schematics, flow diagrams
- A summary of progress to-date
  - Screenshots, video, or even a demo of your prototype?
- How you intend to meet each Requirement
  - What data will you capture on your demonstration flight to evidence this?
  - Could be outputs, but also flight logs, photos, videos…
- Your plan from start to finish, or at least from now forwards
  - Gantt chart?
- Initial reflections on working practices
  - Plus/Delta review (what's working well/do more of, what are you considering changing)

Logistics:

- A computer with screen will be provided
- You should upload a PowerPoint deck or PDF of slides to the unit Team or bring on USB
- You may also connect your own device via a standard HDMI port if you wish

The PDR does not contribute towards your final grade but is an important opportunity to shape your progress and gain an insight into what assessors will be looking for in your final deliverables.

D5 Final Design Review (50%)

- In-person presentation with accompanying slides
- 15 min presentation + 10 min questions
- Each company member should present at least one slide each

Audience:

- MSc peers
- Flight Lab academics and postgrads
- Wider University colleagues
- External collaborators and guests

Present:

- An overview of your system and how it will meet design requirements
- What to expect in your demonstration flight
- How you worked together effectively to meet your goals

Logistics:

- A computer with screen will be provided
- You should upload a PowerPoint deck or PDF of slides to the unit Team or bring on USB
- You may also connect your own device via a standard HDMI port if you wish

The FDR does not contribute towards your final grade but is an important opportunity to showcase your work to internal and external attendees.

D6 Company Report (50%)

- Submission: Group submission of single PDF including link to public GitHub repository
- Maximum length: 15 pages, excluding cover page, exec summary, introduction, references, and appendices.

To include:

1. **Executive summary**
   Single page self-contained summary.
2. **Introduction**
   A summary of the context and background to the project, Company description including member bios and roles, and brief description of member contributions.
3. **Design rationale**
   Justification of design decisions including STEEPLE considerations.
4. **System description**
   The architecture of your final system. Flow charts, schematics, images recommended.
5. **Requirements verification**
   Process and evidence of satisfaction of requirements. Descriptions, images, technical details.
6. **Evaluation**
   Plus/delta review of system technical performance[2].
7. **References**
   As appropriate to cite research/design/code sources and to justify design decisions.
8. **Appendices**
   Further detail e.g. additional schematics, trade-off studies, etc. for completeness and portfolio purposes. Appendices will not be assessed – include excerpts and summaries in main sections.

Assessment will be against relevant Level 7 criteria[3], via the tailored rubric in Appendix A.

[2] This section should focus on things that contributed to technical success. Individual reflections on team working and self-development will be undertaken in D7.
[3] https://www.bristol.ac.uk/academic-quality/assessment/regulations-and-code-of-practice-for-taught-programmes/marking-criteria/

D7 Individual Reflective Report

DRAFT – to be updated to include AHEP4 criteria to be evidenced.

- Submission: Individual submission of single PDF.
- Maximum length: 10 pages, excluding cover page and appendices.

Assessment will be against relevant Level 7 criteria[4], via the tailored rubric in Appendix B.

[4] https://www.bristol.ac.uk/academic-quality/assessment/regulations-and-code-of-practice-for-taught-programmes/marking-criteria/

Clarifications, amendments and errata

Release	Date	Sections	Changes	Author
1.0	26 Nov 2025	All	Initial release	S Bullock

Appendix A: AENGM0074 **D6 Company Report** assessment rubric, benchmarked against [University marking criteria – level 7](#).

Marks	**0, 7, 15, 22, 29, 35**	**42, 45, 48**	**52, 55, 58**	**62, 65, 68**	**72, 75, 78**	**83, 94, 100**
Specialist skills and problem-solving **40%**	Does not show ability to identify key aspects of complex problems or to use appropriate skills and resources to address them.	Little evidence of awareness of key aspects, weak deployment of appropriate tools and techniques.	Limited ability to identify key aspects and use appropriate resources to address them. Some understanding of technologies and approaches demonstrated.	Key aspects of task and challenges identified. Appropriate technologies and approaches selected and implemented effectively.	Key aspects of task and challenges identified. Appropriate technologies and approaches selected and implemented effectively, showing initiative and autonomy.	Confident and comprehensive identification of task and challenge aspects. Appropriate technologies and approaches selected and implemented highly effectively, showing initiative, autonomy, and creativity.
Decision making **40%**	No evidence shown of ability to make decisions in complex and unpredictable circumstances.	Little evidence shown of ability to make decisions in complex and unpredictable circumstances.	Shows limited ability to make decisions in complex and unpredictable circumstances.	Confident in adapting to changing and unfamiliar or challenging circumstances and making evidence-based decisions.	Confident in adapting to changing and unfamiliar/challenging circumstances and making effective, evidence-based decisions.	Shows confidence and creativity in adapting to changing and unfamiliar/challenging circumstances.
Communication **20%**	Very limited awareness of effective communication in reporting and presentation	Limited awareness of effective communication in reporting and presentation.	Some awareness of effective communication in reporting and presentation.	Reporting and presentation are effective, making use of appropriate techniques and resources.	Reporting and presentation are effective, making use of appropriate techniques and resources.	Reporting and presentation are effective, engaging, and professional, making use of innovative techniques and resources.

Appendix B: AENGM0074 **D7 Individual Reflective Report** assessment rubric, benchmarked against University marking criteria – level 7.

Marks	**0, 7, 15, 22, 29, 35**	**42, 45, 48**	**52, 55, 58**	**62, 65, 68**	**72, 75, 78**	**83, 94, 100**
Teamwork	Little or no demonstration of ability to work within a team setting.	Shows limited ability to work within a team setting.	Shows ability to work with others and contribute productively as a member of a team.	Works effectively within a team, recognising the value and contributions of others. Able to manage conflict.	Consistently demonstrates effective teamworking and leadership skills and able to ensure teams work effectively to meet their obligations and goals. Able to manage conflict.	Shows outstanding ability to work and lead a team with creativity and flexibility that is responsive to group members' interests and the obligations and goals of the team. Able to manage conflict.
Self-management	Shows very limited evidence of self-organisational skills and behaviours and ability to meet deadlines.	Shows limited evidence of self-organisational skills and behaviours and ability to meet deadlines.	Shows some evidence of self-organisational skills and behaviours. Able to complete most tasks by deadlines.	Demonstrates good self-organisational skills and behaviours. Has a professional attitude to completing tasks.	Works autonomously demonstrating very good self-organisational skills and behaviours. Has a professional attitude to completing tasks.	Works autonomously demonstrating outstanding self-organisational skills and behaviours. Has a professional attitude to completing all tasks.
Insight	Shows very limited awareness of own strengths and weaknesses.	Displays limited awareness of own strengths or weaknesses.	Shows some ability to identify own strengths and weaknesses. Some evidence of capacity to plan self-development to improve practical and professional skills.	Confident in self-reflection and expressing own strengths and weaknesses and able to take a proactive approach to self-development to improve practical and professional skills.	Demonstrates ability to work autonomously and assess own strengths and weaknesses. Demonstrates ability to identify and implement an effective programme of self-development to improve practical and professional skills.	Shows confidence in working autonomously and setting own goals. Can assess own strengths and weaknesses. Able to identify and implement an effective programme of self-development to improve practical and professional skills. Can provide effective feedback to others to aid their self-development.